ANALPAN

DESCRIPTION:
ANALPAN INJECTION 75mg / 3ml: A clear, colourless to almost colourless liquid in ampoule for IV infusion or I/M injection.
ANALPAN INJECTION 25mg / ml: A clear, colourless to almost colourless liquid in vial for IV infusion or I/M injection.

COMPOSITION:
Each ml contains 25mg/ml Diclofenac Sodium.

PHARMACODYNAMICS:
Diclofenac Sodium is structurally similar to both the phenyl-alkanoic acid series (e.g. Ibuprofen, Alclofenac) and the anthranilic acid series (e.g. Flufenamic acid, Mefenamic Acid). Diclofenac Sodium, a non-steroidal compound, exhibits pronounced antirheumatic, anti-inflammatory, analgesic, and antipyretic properties. As with other NSAIDS, its mode of action is not known; however, its ability to inhibit prostaglandin synthesis may be involved in the anti-inflammatory effect. Analpan injection contains the sodium salt of diclofenac, a non-steroidal substance with pronounced antirheumatic, anti-inflammatory, analgesic and antipyretic effects.

PHARMACOKINETICS:
Distribution: Mean peak plasma concentration of 2.5µg/ml (8µmol/L) are reached approximately 20 min after IM injection of 75mg diclofenac. Plasma concentrations are in linear proportion to the size of the dose. The plasma concentrations observed in children after administration of equivalent doses (mg/kg body weight) are similar to those reached in adults. Pharmacokinetics behavior does not change after repeated administration. No accumulation occurs if the recommended dosage intervals are observed. 99.7% of diclofenac is bound to serum proteins, mainly to albumin (99.4%). Diclofenac enters the synovial fluid, where peak concentrations are measured 2-4 hours after peak plasma levels have been reached. The apparent elimination half-life from the synovial fluid is 3-6 hours. This means that concentrations of active substance in the synovial fluid are already higher than plasma concentrations 4-6 hours after administration and they remain higher for up to 12 hours.
Metabolism: Biotransformation takes place by glucuronidation partly of the intact molecule, but mainly after single and multiple hydroxylation.
Elimination: The active substance is eliminated from the plasma with a systemic clearance of 263±56 ml/min (±SD). The terminal half-life is 1-2 hrs. Approximately 60% of the dose administered is excreted via the kidneys in the form of metabolites, <1% is excreted as unchanged substance. The rest of the dose is eliminated in the form of metabolites through the bile in the faeces.
Kinetics in Special Clinical Situations: No relevant age dependent differences in absorption, metabolism and excretion have been observed.
In patients with impaired renal function, no accumulation of the unchanged active substance can be inferred from the single dose kinetics under the usual dosage schedule. At a creatinine clearance of <10ml/min, the theoretical steady-state plasma levels of metabolites are about 4 times higher than in normal subjects. The metabolite is nevertheless ultimately cleared through the bile. In patients with impaired hepatic function (chronic hepatitis), non-decompensated liver cirrhosis, the kinetics and metabolism of diclofenac are the same in patients without liver disease.

INDICATIONS:
1. Inflammatory & degenerative forms of rheumatism
2. Rheumatoid arthritis
3. Ankylosing spondylitis
4. Osteoarthritis, spondylarthritis
5. Non-articular rheumatism
6. Painful inflammatory conditions of non-rheumatic origin

RECOMMENDED DOSAGE:
Adult: ***Intravenous Infusion*:** For the treatment of moderate to severe post-operative pain, 75mg should be infused continuously over a period of 30 minutes to 2 hours. If necessary, treatment may be repeated after 4-6 hours, not exceeding 150mg within any period of 24 hours. For the prevention of post-operative pain, a loading dose of 25mg-50mg should be infused after surgery over 15 minutes to 1 hour, followed by a continuous infusion of approx. 5mg per hour up to a maximum daily dosage of 150mg. ***Intramuscular injection***: One ampoule once (or in severe cases twice) daily intramuscularly by deep intragluteal injection into the upper outer quadrant. If two injections daily are required it is advised that the alternative buttock be used for the second injection. Alternatively, one ampoule of 75mg can be combined with other dosage forms (tablets or suppositories) up to the maximum daily dosage of 150mg.
Note: After assessing the risk/ benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used. The higher dose is more often required in patients with rheumatoid arthritis than in patients with osteoarthritis. Diclofenac Sodium Injection should not be given for more than 2 days. If necessary, treatment can be continued with tablets or suppositories. Diclofenac Injection also should not be administered by intravenous bolus injection. As a general recommendation, the dose should be individually adjusted. Adverse effects may be minimized by using the lowest effective dose for the shortest duration necessary to control symptoms (see section WARNINGS AND PRECAUTIONS).
Established cardiovascular disease or significant cardiovascular risk factors: Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤100mg daily if treated for more than 4 weeks (see section WARNINGS AND PRECAUTIONS).
Note: As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

ROUTE OF ADMINISTRATION: For Intravenous Infusion or Intramuscular (I.M) Injection.

CONTRAINDICATIONS:
1. Known allergy to Diclofenac Sodium
2. Cross sensitivity has been demonstrated between diclofenac, aspirin and naproxen. Diclofenac should not be given to aspirin sensitive patients. Sensitivity phenomena include asthma, rhinitis and urticaria.
3. Active gastrointestinal bleeding (eg. peptic ulcer perforation) recurrent dyspepsia or other gastrointestinal pathology.
4. Severe hepatic impairment
5. Pregnancy (see section Use in Pregnancy)
6. Sever cardiac failure (see section WARNINGS AND PRECAUTIONS)

WARNINGS & PRECAUTIONS:
History of gastrointestinal haemorrhage or ulcer: Diclofenac should be used with caution in these patients. Patients with history of dyspepsia or other gastrointestinal disorder such as Crohn's disease and ulcerative colitis or with pre-existing dyshaemopoiesis or disorders of blood coagulation, as well as those with severe hepatic or renal disease, should be kept under close surveillance. In elderly patients who are generally more prone to side effects, particular caution should be exercised. If peptic ulcer or gastrointestinal bleeding occurs during treatment with diclofenac, administration of the drug must cease immediately.
Blood effects: A slight reduction in haemoglobin has been observed in some patients during long-term therapy with diclofenac. On rare occasions, blood dyscrasias have been reported. It is advisable to perform blood counts, at intervals, in patients receiving long-term therapy.
Cardiovascular effects: Treatment with NSAIDs including diclofenac, particularly at high dose and in long term, maybe associated with an increased risk of serious cardiovascular thrombotic events (including myocardial infarction and stroke).
Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤100mg daily when treatment continues for more than 4 weeks. As the cardiovascular risks of diclofenac may increase with dose and duration of exposure, the lowest effective daily dose should be used for the shortest duration possible. The patient's need for symptomatic relief and response to therapy should be re-evaluated periodically, especially when treatment continues for more than 4 weeks.
Patients should remain alert for the signs and symptoms for serious arteriothrombotic events (e.g. chest pain, shortness of breath, weakness, slurring of speech), which can occur without warnings. Patients should be instructed to see a physician immediately in case of such an event.
There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSAID use.
Hypertension: NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking antihypertensive with NSAIDs may have an impaired antihypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.
Heart Failure: Fluid retention and oedema have been observed in some patients taking NSAIDs, there caution is advised in patients with fluid retention or heart failure.
Severe Skin Reactions: Severe cutaneous reactions, including Stevens-Johnson syndrome and toxic epidermal necrolysis (Lyell's syndrome), have been reported with diclofenac sodium. Patients treated with diclofenac sodium should be closely monitored for signs of hypersensitivity reactions. Discontinue diclofenac sodium immediately if rash occurs.
Adverse effects: Dermatological: Please see [SIDE EFFECTS/ADVERSE EFFECTS].
Gastrointestinal effects: NSAIDS, including diclofenac, may be associated with increased risk of gastrointestinal anastomotic leak. Close medical surveillance and caution are recommended when using diclofenac after gastrointestinal surgery.
WARNING FOR SODIUM METABISULPHITE: This preparation contains sodium metabisulphite, a sulphite that may cause allergic-type reactions including anaphylactic symptoms and life-threatening or less severe asthmatic episodes in certain susceptible people. The overall prevalence of sulphite sensitivity in the general population is unknown and probably low. Sulphite sensitivity is seen more frequently in asthmatic than non-asthmatic people.
Use in Children: Diclofenac is not recommended for use in children as safety and efficacy in this age group has not been established.
WARNING RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID: Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients

treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms. Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events. When gastrointestinal bleeding or ulcerations occur in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctor should warn patient about signs and symptoms of serious gastrointestinal toxicity. The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.

Effects on Ability to Drive and Use Machine: Patients who experience visual disturbances, dizziness, vertigo, somnolence, central nervous system disturbances, drowsiness or fatigue while taking NSAIDs should avoid driving or using machines.

INTERACTION WITH OTHER MEDICATIONS:

Lithium & Digoxin: Diclofenac sodium may increase plasma concentrations of lithium and digoxin.

Anticoagulants: Clinical investigation do not appear to indicate that Diclofenac Sodium affects the activity of anticoagulants (eg: warfarin) but there have been isolated reports of an increased risk of haemorrhage with the combined use of these agents. Close monitoring is therefore recommended.

Antidiabetic: Clinical studies have shown that Diclofenac Sodium can be given together with oral antidiabetic agents without influencing their clinical effect. However, there have been isolated reports of hypoglycaemic and hyperglycaemic effects, which have required adjustments to the dosage of hypoglycaemic agents.

Cyclosporin: Cyclosporin nephrotoxicity may be increased by the effect of non-steroidal anti-inflammatory drugs on renal prostaglandins.

Methotrexate: Caution should be exercised if NSAIDs and methotrexate are administered within 24 hours of each other since NSAIDs may increase methotrexate plasma levels resulting in increased toxicity. **NSAIDS:** Concomitant administration of systemic NSAIDS may increase the occurrence of side effects. Concurrent treatment with aspirin lowers the plasma concentration, peak plasma levels and AUC values of diclofenac. The use of both drugs concurrently is not recommended.

Quinolone Antibacterials: There have been isolated reports of convulsions, which may have been due to concomitant use of quinolones and NSAIDS.

Phenytoin: When using phenytoin concomitantly with diclofenac, monitoring of phenytoin plasma concentrations is recommended due to an expected increase in exposure to phenytoin

Potent CYP2C9 inhibitors: Caution is recommended when co-prescribing diclofenac with potent CYP2C9 inhibitors (such as voriconazole), which could result in a significant increase in peak plasma concentrations and exposure to diclofenac due to inhibition of diclofenac metabolism.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: NSAIDS inhibit prostaglandin synthesis and, when given during the latter part of pregnancy, may cause closure of the fetal ductus arteriosus, fetal renal impairment, inhibition of platelet aggregation, and delay labour and birth. Continuous treatment with NSAIDS during the last trimester of pregnancy should only be given on sound indications. During the last few days before expected birth, agents with an inhibitory effect on prostaglandin synthesis should be avoided. Safety of diclofenac sodium in pregnancy has not been established. Therefore, Analpan should not be used in pregnant women or those likely to become pregnant unless the expected benefits are likely to outweigh any possible risk.

Use in Lactation: Following oral doses of 50mg administered every 8 hours, the active substance, diclofenac, passes into breast milk. As with other drugs that are excreted in milk, Analpan is not recommended for use in nursing women.

SIDE EFFECTS/ ADVERSE EFFECTS:

Gastrointestinal Tract:

Common: Epigastric pain, other gastrointestinal disorders such as nausea, vomiting, diarrhoea, abdominal cramps, dyspepsia, flatulence, anorexia.

Rare: Gastrointestinal bleeding, haematemesis, melaena, gastric or intestinal ulcer with or without bleeding or perforation, bloody diarrhoea.

Very Rare: Lower gut disorders such as non-specific haemorrhagic colitis and exacerbation of ulcerative colitis or Crohn's proctocolitis; aphthous stomatitis, glossitis, oesophageal lesions, constipation, diaphragm-like intestinal strictures, pancreatitis.

Central (and peripheral) Nervous System:

Common: headache, dizziness, or vertigo.

Rare: Somnolence, tiredness.

Very Rare: disturbances of sensation, including paraesthesia, memory disturbance, disorientation, disturbance of vision (blurred vision, diplopia), impaired hearing; tinnitus, insomnia, irritability, convulsions, depression, anxiety, nightmares, tremor, psychotic reactions, taste alteration disorders, myoclonic encephalopathy (described in two patients), aseptic meningitis.

Cardiac disorders:

*Uncommon**: Myocardial infarction, cardiac failure, palpitations, chest pain. [*The frequency reflects data from long-term treatment with a high dose (150mg/day)]

Kounis syndrome: Frequency "not known"

Arteriothrombotic events:

Meta-analysis and pharmacoepidemiological data point towards an increased risk of arteriothrombotic events (for example myocardial infarction) associated with the use of diclofenac, particularly at a high dose (150mg daily) and during long- term treatment (see section WARNINGS AND PRECAUTIONS).

Dermatological:

Occasionally: rashes or skin eruptions, cases of hair loss, bullous eruptions, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (Lyell's syndrome), and photosensitivity reactions have been reported.

Rare: urticaria.

Very Rare: eczema, erythroderma (exfoliative dermatitis), purpura, including allergic purpura.

Urogenital System: *Very Rare:* acute renal failure, urinary abnormalities such as haematuria, proteinuria, interstitial nephritis, nephrotic syndrome, papillary necrosis.

Liver:

Common: elevation of serum aminotransferase enzymes (ALT, AST)

Rare: hepatitis with or without jaundice, liver disorder.

Very Rare: fulminant hepatitis, hepatic necrosis, hepatic failure.

Blood: *Very Rare:* thrombocytopenia, leukopenia, anaemia (haemolytic anaemia, aplastic anaemia), agranulocytosis, positive Coombs' test.

Hypersensitivity: *Less than 1%:* hypersensitivity reactions such as bronchospasm, anaphylactic/anaphylactoid systemic reactions including hypotension. *In isolated cases:* vasculitis, pneumonitis.

Eye and Ear:

Common: Vertigo.

Very Rare: Visual disturbance, vision blurred, diplopia, Tinnitus, hearing impaired

Others: *In isolated cases:* impotence (association with diclofenac sodium intake is doubtful), palpitation, chest pain, hypertension, congestive heart failure.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: There have been no recorded cases of overdosage with Diclofenac. In the event of acute poisoning, the following symptoms and signs might be encountered.

Nervous System: Headache, motor restless, muscular twitching, hyperexcitability, ataxia, dizziness.

Convulsions: especially in small children.

Gastrointestinal: Epigastric pain, nausea, vomiting, haemorrhage, diarrhoea, acute ulceration of stomach or intestine.

Liver: Disorder of hepatic function.

Kidneys: Oliguria.

Laboratory: Raised SGOT/SGPT, Hyperbilirubinaemia.

Treatment: There is no specific antidote. General measures should be taken to reduce the amount of Diclofenac absorbed (e.g. induction of vomiting, gastric lavage, or the use of activated charcoal) and to increase the rate of eliminations (e.g. dialysis, especially indicated in the presence of renal failure). Convulsion may be controlled by the use of diazepam or phenobarbitone. Surveillance should also include monitoring of the blood picture, serum electrolytes, liver and kidney function, and examination of stools (for occult blood).

INCOMPATIBILITIES:
This product used for im or iv as an infusion should not be mixed with other injection solutions.

STORAGE CONDITIONS:
Store below 30°C. Protect from light. Keep out of reach of children. Jauhkan daripada kanak-kanak

SHELF LIFE:
Please refer to outer package.

PACK SIZE:

Ampoules : 3ml x 10 ampoules and 3ml x 100 ampoules.
Vials : 10ml x 25 vials

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA

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